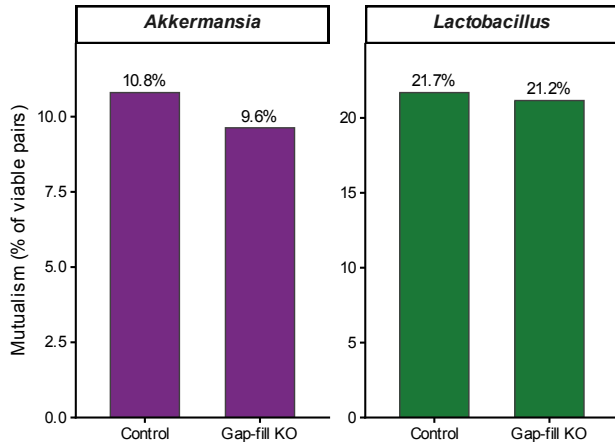


A. Overall mutualism is nearly unchanged

After blocking dormant no-GPR internal reactions
(transporters kept; monoculture growth unchanged)



B. $\geq 89\%$ of mutualistic pairs keep their classification

Akkermansia n=1,718; *Lactobacillus* n=5,753
control-mutualistic viable pairs

→ neutral / commensal → competition Retained mutualism

